Energy Research & Social Science 120 (2025) 103917

Contents lists available at ScienceDirect

Energy Research & Social Science

journal homepage: www.elsevier.com/locate/erss

Original research article

Leveraging geospatial data to evaluate women's employment opportunities in the renewable energy sector

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ARTICLE INFO

Keywords:
Spatial analysis
Employment
Renewable energy
Women
Access
Jobs

ABSTRACT

The Renewable Energy (RE) sector has grown significantly in response to the global climate crisis, yet women's participation remains disproportionately low, with many confined to lower-paid, non-technical roles. This persistent gender disparity could limit the sector's potential unless diversity is actively promoted. Traditional studies on women's employment in RE often overlook critical spatial factors, such as access to public transportation, safety, and neighborhood walkability, which are vital for understanding women's job access. To address these gaps, a Geospatial Women's Employment Analytical Framework (GeoWEAF) was developed, identifying 15 key spatial factors across three dimensions: Contextual, Accessibility, and Place-Characterization. These dimensions are crucial for evaluating whether an environment supports women's access to employment in the RE sector. The GeoWEAF was piloted in Saint Lucia to assess its practical application and effectiveness. It was designed as a decision-making tool for policymakers, offering data-driven insights that enable evidence-based decisions accounting for geographic disparities. By integrating spatial indicators into the evaluation of women's employment opportunities, the GeoWEAF provides a comprehensive understanding of the spatial dynamics shaping women's employment prospects, thereby facilitating targeted interventions to promote gender equity in the workforce.

1. Introduction

Amid the pressing global urge to combat climate change, the Renewable Energy (RE) sector has experienced significant expansion [1]. However, this growth has not been universally beneficial; women's participation in formal employment within this rapidly developing field significantly trails that of their male counterparts [2–5]. Furthermore, women in the sector are predominantly employed in lower-paid administrative and non-technical roles, rather than in higher-paid positions that require advanced training and experience [3,6,7]. Consequently, there is growing concern that without prioritizing gender diversity, the RE transition could perpetuate and even exacerbate gender inequality rather than alleviate it [4,5,8]. Despite evidence suggesting that diversity fosters innovation and creativity, there has been minimal focus on promoting the inclusion of women within the RE workforce [5,9].

To ensure a truly sustainable energy transition, it is essential to increase women's participation in the sector, recognizing their potential to contribute to sustainable development and innovative energy solutions [2,4,10]. To this end, conducting comprehensive studies that assess the barriers women face in accessing employment opportunities in the sector is essential [11], especially in low- and middle-income countries where socio-economic constraints, cultural norms, and infrastructural challenges can exacerbate these barriers [11,12]. Nonetheless, traditional approaches to evaluating women's job access in the RE sector frequently overlook critical spatial data that could substantially influence women's employment opportunities [13,14]. Omitting considerations such as the geographic distribution of environmental characteristics, resources, and services—including public transportation, childcare services, and safety and harassment concerns in specific locations—undermines a comprehensive analysis of the factors that affect women's entry and retention in the workforce [15].

To address this gap, the present research has two main objectives. The first is to develop an analytical framework that identifies key

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https://doi.org/10.1016/j.erss.2024.103917
Received 25 September 2024; Received in revised form 17 December 2024; Accepted 21 December 2024
Available online 30 December 2024
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location-specific factors influencing women's access to employment opportunities in the RE sector, specifically in low- and middle-income countries. The second is to apply this framework in a selected country, using spatial analyses to characterize regions by how conducive they are for women to access job opportunities, and to highlight the spatial factors that drive these regional differences. In order to achieve these two objectives, the research will explore two corresponding key questions:

1) What spatial factors within a given location influence women's access to employment opportunities in the RE sector?
2) How do these factors contribute to regional variations in women's access to jobs in the RE sector within the selected country?

This analytical framework will contribute to the existing literature on women's barriers to accessing jobs in the RE sector by incorporating spatial factors, not traditionally considered, representing a pioneering effort to understand the geographical dimensions of the gender gap. By incorporating spatial considerations into the assessment of women's job access, this research aims to overcome the limitations of existing frameworks and provide a more comprehensive, nuanced, and visually engaging understanding of the barriers and opportunities for women to access employment in low- and middle-income countries.

2. Background

Evidence suggests that women perceive and experience spaces in ways that differ significantly from men [16–19]. Consequently, geography plays a crucial role in shaping women's mobility patterns and behavioral choices, which in turn affects their job preferences and opportunities [20,21]. For instance, the availability and characteristics of services, amenities, goods, and industries in areas where women reside creates vital opportunities for their inclusion and retention in the workforce [21,22]. As observed by Sorenson and Dahl [23], if locations are more suitable for men's employment than for women's, this mismatch will result in lower wages for women in that region. In light of this, understanding how geography and spatial features influence women's career progression and employment opportunities is essential for developing policies and programs aimed at reducing the gender gap.

For example, Rosenbloom [24] argues that understanding women's travel patterns is crucial for uncovering the barriers that certain locations may pose to their job access. Consistent with this argument, evidence worldwide has shown that women typically travel shorter distances and spend less time commuting compared to men [21,25]. This difference in travel patterns is largely attributed to the nature of women's trips, which are characterized by multiple stops [26–29]. In contrast, men's trips tend to be more linear, often involving direct routes to a single destination [30]. The multi-stop nature of women's trips is often attributed to the imbalance in household responsibilities, including the frequent need to accompany children or elderly family members and to perform errands such as grocery shopping [21,27,31]. Additionally, recent studies have linked this tendency to economic disparity, which often makes women more reliant on public transportation or active transport [32]. These differences underscore the distinct daily logistical challenges and constraints faced by women, reflecting broader societal roles and expectations that undoubtedly influence their ability to access employment opportunities. This imbalance is even more pronounced in developing contexts, where limited infrastructure, inadequate transportation systems, and deeply entrenched societal norms exacerbate these challenges [33,34].

Furthermore, some studies even suggest that women are more likely than men to accept lower wages in exchange for shorter commutes, enabling them to fulfill household responsibilities [35]. This tendency contributes to an increased wage gap and fewer employment opportunities for women, often restricting them to jobs closer to home [36]. Similarly, when childcare options are limited, costly, or of poor quality, women are less likely to work and more likely to hold low-quality jobs [37]. As a result, the scarcity of essential services such as reliable public transportation, nearby childcare facilities, grocery stores, and health clinics often limits women's access to high-paying or full-time employment opportunities [15,21]. In turn, women are more likely to choose workplaces that require less commute time and where they can accommodate their needs as primary caregivers in the home [38].

Additionally, women often exhibit greater concerns about safety compared to men [39–41]. For instance, research, such as Ode Sang et al.'s [42] study in Gothenburg, Sweden, and Carli's [43] analysis across 40 major cities in China, suggest that safety significantly influences women's mobility and use of public spaces. Similarly, Mayen Huerta and Utomo [40] highlight that women in Latin America often experience heightened feelings of insecurity when navigating public spaces. Valentine [44] described this phenomenon as the "geography of fear," highlighting how feelings of vulnerability shape women's choices, mobility, and engagement with public spaces. The geography of fear compels women to adjust their behavior and travel patterns to avoid spaces that do not meet their specific safety and security needs, thereby influencing their job prospects [45]. This geography of fear is even more pronounced in countries where conflict and violence are persistent [46], intensifying the barriers women face in participating fully in the workforce.

Evidence from studies conducted worldwide has linked characteristics such as lighting, cleanliness, maintenance, order, comfort, and aesthetics of public spaces to women's safety perceptions, with women tending to prioritize these attributes more than men [19,47,48]. For example, survey results from a study in Carmona, Spain, revealed that women place a higher value on the quality and safety features of public spaces [49]. Likewise, evidence from Londrina, Brazil, found that women often regulated their activities and limited their use of public spaces to certain times of the day due to perceived safety concerns, especially in poorly lit and poorly maintained areas [50]. Accordingly, women are more likely to utilize areas perceived as safer and of higher quality [49].

Given the significant impact of geography on women's mobility patterns and behavioral choices, a more nuanced understanding of how geography affects job opportunities requires the inclusion of spatial considerations in the analysis. This is particularly important when analyzing women's employment and business opportunities in the RE sector, as its potential production and diffusion have a spatial dimension [51,52]. Understanding the reasons behind the mismatch between potential RE sites and environments that encourage greater participation by women is crucial for bringing more women into leadership roles in the sector, thus ensuring more balanced opportunities. This is particularly important in developing contexts, where poor infrastructure, safety concerns, and deep-rooted cultural constraints often hinder women's access to remote job sites and professional development opportunities, negatively affecting their economic prospects [53–55]. Reducing the gender gap in these regions is crucial to ensuring a just and inclusive energy transition, empowering women to participate fully and equitably in the growing RE sector [56].

From a policy standpoint, integrating geospatial data with sex-disaggregated economic and social statistics can significantly enhance the formulation of gender-sensitive policies and programs [57,58]. Such integration is crucial for accelerating RE deployment and fostering low-carbon economic growth [59]. Furthermore, incorporating spatial indicators into an employment assessment framework can significantly improve the evaluation of opportunities by providing a comprehensive understanding of the spatial dynamics that shape employment landscapes [21]. This approach can enable the development of targeted interventions to promote gender equality in the workforce.

3. Case study: Saint Lucia

For the practical application of the analytical framework (objective 2), St. Lucia will be used as a case study for three key reasons. First, as a

small island developing state (SIDS), St. Lucia is highly susceptible to the impacts of climate change, including sea-level rise, increased storm intensity, and other related hazards that could severely damage its infrastructure and economy [60–62]. Moreover, like many other middle-income SIDS, St. Lucia relies heavily on imported fossil fuels and is consequently vulnerable to the price volatility and geopolitical issues associated with these energy sources [63]. Therefore, transitioning to RE and ensuring that women are actively involved in the planning and implementation processes of this transition are both crucial for the country's resilience and sustainable economic growth. This transition is urgent to reduce the country's carbon footprint, enhance energy security, and mitigate the impact of volatile fossil fuel costs [63,64].

Second, Saint Lucia was chosen as a case study due to its challenging data landscape, allowing us to test the analytical framework's applicability in countries with limited data availability. This aims to demonstrate that even in countries where comprehensive public geospatial data is not available, simple analyses can still be performed to better understand the configuration of the environment. Moreover, by selecting a country where many features remain unmapped, we can demonstrate the framework's effectiveness and applicability to other low- and middle-income countries, including those that frequently encounter data completeness challenges.

Third, Saint Lucian society has traditionally adhered to patriarchal norms, where women are expected to manage household responsibilities and caregiving roles [65,66]. These expectations can hinder women's ability to pursue full-time employment or professional careers, especially when jobs are far from home or essential services supporting their caregiving duties are unavailable. Therefore, understanding the specific factors needed in areas where RE jobs are available is crucial to increasing women's participation in the field. Moreover, in Saint Lucia, the tourism sector dominates, offering women job opportunities primarily in hospitality and service industries [67]. However, these roles are often characterized by low wages, seasonal work, and limited potential for advancement [68]. To reduce the gender wage gap, it is essential to improve the conditions that enable women to access and thrive in higher paying, more stable careers. Thus, a study of this nature within this context becomes highly relevant.

Notably, in recent years, Saint Lucia has commenced its first large-scale clean energy project, a solar plant located near the international airport [69]. With a peak electricity demand of approximately 60 MW, this project represents a significant advancement for the island's energy infrastructure [69]. As such, this research will focus on analyzing the spatial barriers that limit job prospects around this major project, since the smaller solar power plants on the island provide fewer employment opportunities. As is the case in many Caribbean countries, women in Saint Lucia are scarcely involved in efforts to advance the energy transition, which perpetuates existing inequalities [4]. The ongoing energy transition offers a valuable opportunity to advance justice and address gender equality issues in the island [4]. If gender-blindness persists within energy systems, achieving these objectives may take considerably longer.

4. Methodology

To address objective 1, we developed the Geospatial Women's Employment Analytical Framework (GeoWEAF), which identifies key spatial factors that foster a supportive environment for women to secure formal employment or business opportunities in the renewable energy sector. This framework was informed by an extensive literature review exploring the intersection of geography and women's employment. Notably, we focused on labor markets in the fields of Science, Technology, Engineering, and Mathematics (STEM), as this research aims not only to assess the environment for women to access job opportunities but also to secure higher-paying positions that require advanced competencies. Additionally, we consulted three World Bank experts—a geospatial expert, a gender labor markets expert, and an energy sector expert—to ensure the selected factors were appropriate and accurately reflected the spatial barriers faced by women in the RE sector.

It is important to emphasize that the GeoWEAF was specifically developed with low- and middle-income contexts in mind, as the barriers women face in these settings are distinct from those in higher-income environments. In conducting the review, we identified relevant studies in low- and middle-income countries using targeted search terms and Boolean operators such as "spatial analysis AND women's employment," "geography AND renewable energy sector," "gender equity AND labor market access," "spatial inequality AND employment opportunities," "women's economic empowerment AND spatial accessibility," and "spatial distribution AND labor market participation" in databases like Google Scholar, Scopus, and Web of Science. These search strategies ensured that we captured a wide range of literature addressing the spatial dimensions of women's employment. The literature review followed a saturation approach, meaning that once key themes began to recur without the emergence of new insights, we deemed the review complete [70,71]. At that point, we categorized and classified all identified themes for further analysis. We identified 15 key spatially varying factors, which we categorized into three distinct dimensions based on their nature (see Fig. 1).

Factors in the Contextual Dimension (CD) of the GeoWEAF pertain to the laws and policies that influence workplace gender discrimination, financial autonomy, and overall gender empowerment [72–76]. Factors in the Accessibility Dimension (AD) relate to women's daily mobility and travel patterns. These factors consider the proximity from women's homes to essential services, which is a crucial indicator of accessibility [77–79]. Finally, factors in the Place-Characterization Dimension (PD) pertain to the attributes of geographical locations or environments. These include walkability, safety, and vulnerability to natural disasters, all of which significantly impact women's choices and freedom [45,80].

4.1. Factors and rationale

The following section outlines the GeoWEAF factors within each dimension, providing a brief rationale for their inclusion in the framework.

4.2. Contextual dimension

4.2.1. Workplace discrimination

Gender biases and stereotypes in recruitment and workplace cultures often hinder women's career progression in male-dominated fields, such as the RE sector, by perpetuating gender prejudices [81,82]. Such discrimination can manifest in various forms, from hiring practices to professional development opportunities, impacting women's employment outcomes and job satisfaction [83]. Scholars like Emmons Allison et al. [3] argue that, particularly in the RE sector, the societal underestimation of women's competencies in technical fields, coupled with women's justified expectations of mistreatment and harassment, present formidable obstacles to entry.

4.2.2. Regulatory frameworks (RF)

Genuine equal opportunity relies on the effective legal protection of women's employment rights and interests [84]. As such, variations in childcare support, parental leave, and other gender equality policies across regions can either enable or restrict women's access to the workforce [85]. For instance, a study conducted in the U.S. analyzing the impact of different state maternity leave policies on mothers' employment found that mothers in states with expanded access to job-protected leave were less likely to resign from their jobs [86]. Similarly, a study examining the impact of job protection policies on gender differences in participation and wages found that such policies significantly increase the participation of mothers with children under the age of 6 [87]. These findings underscore the critical importance of robust and consistent employment protection policies in ensuring women's

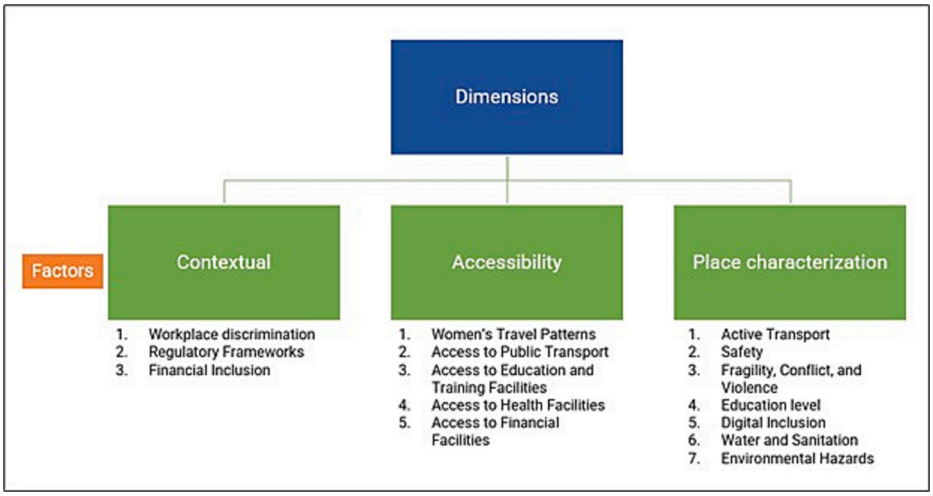

Fig. 1. GeoWEAF factors and dimensions.

sustained participation in the workforce and advancing gender equality [37,88].

4.2.3. Financial inclusion

Women's right to financial resources, encompassing loans, credit, and banking services, is fundamental for initiating businesses and participating in entrepreneurial ventures [2,89]. Disparities in access to finance can limit women's economic empowerment and their ability to invest in training or business opportunities [6,72]. This issue is particularly pronounced in sectors that often require substantial initial investments, such as the RE sector [3]. As such, laws ensuring equal access to credit and banking services are critical in addressing disparities and empowering women to fully participate in economic opportunities.

4.2.4. Accessibility dimension

4.2.4.1. Women's travel patterns (WTP). This factor addresses the unique travel behaviors and constraints faced by women, which are often influenced by their roles as caregivers and household responsibilities [27,31]. Women's travel involves multi-stop journeys for family and household-related errands, which are usually conducted during off-peak hours [26–29]. These errands often encompass visits to grocery stores, childcare services, parks, and pharmacies [90]. Therefore, the proximity of these services is crucial for effectively accommodating the daily needs of women.

4.2.4.2. Access to public transport. Particularly in low- and middle-income countries, due to reduced car ownership, the reliance on public transportation is more significant for women than men, with the availability and quality of such services playing a crucial role in their ability to access jobs, education, and other essential services [15,45,91]. Dobbs [92] argues that the lack of high-quality, safe public transport can result in the social exclusion of women, particularly in terms of accessing employment opportunities outside of their neighborhoods. As such, having walking access to public transportation stops is crucial for increasing women's participation in the labor market [93,94].

4.2.4.3. Access to education and training facilities. Physical access to higher education institutions and training facilities is a crucial factor in determining educational participation [78]. In particular, low-income and ethnic minority women often face significant constraints in their educational choices due to financial or cultural reasons that necessitate them staying at home [78]. Studies such as Frenette [95] in Canada and Matsolo et al. [96] in South Africa have shown that students residing beyond commuting distance are significantly less likely to attend higher education institutions compared to those living within commuting distance. Therefore, this factor assesses the ease with which women can access universities and training facilities, which is crucial for acquiring the qualifications necessary for attaining high-paid technical roles in the RE sector [3,6,7].

4.2.4.4. Access to health facilities. Preventing obstetric complications and maternal deaths requires safe delivery practices, which are facilitated by providing women with greater access to skilled healthcare professionals at medical facilities [97]. Moreover, given women's roles in family healthcare, access to health facilities is more critical for them than men [98,99]. Thus, this factor evaluates the ease with which women can reach healthcare services, which influences their well-being and ability to work.

4.2.4.5. Access to financial facilities. The physical location of financial institutions remains relevant despite the rise of digital financial services. For instance, in their analysis of women's access to credit from rural banks in Ghana, Akudugu et al. [100] observed that the long distances to banks significantly hindered women's ability to apply for credit. Similarly, in a study analyzing financial access in India, Bhanot et al. [101] found that banking facilities were located >3 km away for 67 % of respondents, which significantly limited their ability to apply for credit. Hence, this factor looks at women's proximity to financial services, which is essential for their economic empowerment and ability to engage in business ventures [102,103], including those in the RE sector.

4.2.5. Place-characterization dimension

4.2.5.1. Active transport. In developing contexts, women are more likely than men to rely on active transport, particularly walking, as their primary mode of transportation [45,104]. High walkability and well-connected streets can significantly facilitate women's access to employment by providing safe and efficient travel options [105]. Moreover, street access to a diverse mix of land uses increases the likelihood that women will walk or cycle [104]. On the other hand, poorly maintained, unsafe, or non-walkable environments—characterized by inadequate street lighting, lack of pedestrian paths, absence of ramps, and unsafe crossing points—often deter women's mobility [45]. Thus, considering attributes that hinder walkability or cycling is crucial when assessing the physical barriers women face in accessing jobs.

4.2.5.2. Safety. The safety of public spaces is a major concern for women, impacting their decisions on when to go out and which routes to

take [39,40,106]. Areas perceived as unsafe due to high levels of violence or sexual harassment can deter women from seeking employment or accessing essential services [45,107]. For instance, a study conducted in New Zealand examining mobility patterns found that perceived safety at stations was a significant factor only for female riders, who tended to avoid stations they perceived as unsafe [106]. Lino and Kanashiro [50] argue that women who perceive insecurity as high may be limited in their mobility at a micro level, restricted to their neighborhoods, which has repercussions at a macro level, including women's ability and willingness to work.

4.2.5.3. Fragility, conflict, and violence (FCV). There is no doubt that political violence, conflict, inequality, and displacement are gendered, disproportionately affecting women and girls, and exposing them to higher levels of violence and exclusion [108,109]. FCV makes it harder for women to access services and increases their unpaid caregiving workload, limiting their ability to work outside the home [110]. Furthermore, due to their heightened vulnerability, some women in FCV contexts are unable or restricted from leaving their homes, severely affecting their income-generating possibilities [111]. Therefore, this factor assesses the occurrence and frequency of events that lead to conflict and violence in a specific region.

4.2.5.4. Education. Higher educational attainment, especially in STEM fields, is crucial for accessing technical roles in the RE sector [112]. However, disparities in STEM education between genders often limit women's entry into these technical roles [113,114]. A higher proportion of female STEM graduates in a region has been linked to more favorable societal attitudes toward women pursuing STEM careers [115]. Consequently, due to the correlation between enrollment in STEM fields and gender-related stereotypes and biases [114,116], measuring women's educational attainment in STEM within a region is essential for understanding societal expectations regarding women's career preferences.

4.2.5.5. Digital inclusion. Digital connectivity and access to information and communication technologies are increasingly important for accessing job opportunities, training, and financial services [117,118]. Regions with stable access to the Internet are more conducive to sustaining businesses and jobs, including those requiring technical skills and equipment operation [2]. Furthermore, women with Internet access have a significant advantage in the job market, as it facilitates the search for opportunities that are better suited to their skills and preferences [119]. Digital inclusion also enables women to effectively recruit qualified candidates for their businesses, expand their professional networks, and access remote work opportunities, further enhancing their participation in the economy [120].

4.2.5.6. Water and sanitation. The availability of clean water and sanitation facilities is a fundamental aspect of community health and well-being, affecting women's time allocation and their capacity to engage in employment outside the home [121]. Since women and girls in developing contexts are often tasked with domestic duties, they predominantly bear the responsibility for water collection in areas where water and sanitation are limited [80]. As such, the availability of clean water and sanitation facilities is vital for promoting women's empowerment.

4.2.5.7. Environmental hazards. Environmental and climatic conditions can impact the availability and stability of jobs, especially in sectors sensitive to environmental changes like agriculture and RE [122,123]. In addition, women are 14 times more likely than men to die as a result of natural disasters and are significantly more vulnerable to their effects, including food insecurity [109,124]. Accordingly, any assessment of women's employment must critically consider the heightened risk of environmental hazards, such as droughts, floods, landslides, and fires, acknowledging the profound implications these threats have on women's livelihoods and overall well-being.

4.3. Data and methods

The following section details the selected methods and data used in implementing the GeoWEAF. Importantly, this framework could be applied to various countries and administrative levels, enabling both local and national analyses. To achieve this, applying the same methodology with adjusted boundaries for the area of interest would be sufficient.

4.3.1. Contextual dimension

Since policies and laws are generally applied uniformly across a country, the data used to evaluate the factors in the CD were collected at the national level. The scores for all three CD factors were derived from the World Bank's *Women, Business, and the Law* (WBL) study, which assesses various criteria on a scale from 0 to 100 [125]. These scores were then standardized to a scale of 0 to 5, where 0 represents the lowest score (0) and 5 represents the highest (100).

4.3.1.1. Workplace discrimination. This factor was assessed using the "Workplace" rating (score = 100) from the 2024 WBL report, which evaluates four key criteria to measure workplace discrimination in a country: (1) whether a woman can secure employment on the same terms as a man, (2) whether the law explicitly prohibits employment discrimination based on gender, (3) whether there is legislation addressing sexual harassment in the workplace, and (4) whether there are criminal penalties or civil remedies for such harassment [125].

4.3.1.2. RF. RF was derived from the average of the "Pay" (score = 100) and "Parenthood"(score = 40) ratings. The Pay rating examines the presence of four criteria in a country: (1) whether equal remuneration for work of equal value is mandated by law, (2) whether women are allowed to work at night on the same terms as men, (3) whether women can work in jobs classified as dangerous on the same terms as men, and (4) whether women can work in industrial jobs on the same terms as men [125]. Meanwhile, the Parenthood rating evaluates four additional criteria: (1) whether mothers are provided with at least 14 weeks of paid leave, (2) whether the government funds 100 % of maternity leave benefits, (3) whether fathers are entitled to paid leave, and (4) whether laws protect pregnant workers from dismissal [125].

4.3.1.3. Financial inclusion. Finally, financial inclusion was derived from the "Entrepreneurship" rating (score = 75). This rating examines four key aspects in a country: (1) whether laws prohibit gender-based discrimination in access to credit, (2) whether women have the same rights as men to sign contracts, (3) whether women can register businesses under the same conditions as men, and (4) whether women can open bank accounts with the same ease as men [125].

4.3.2. Accessibility dimension

In general, all AD factors were evaluated using catchment areas calculated through network analyses for walking distances around key facilities in Open Route Service, which relies on the road network data provided by OpenStreetMap (OSM). The travel distance used for walking is based on a speed of **5 km/h**, which corresponds to a standard average walking pace commonly used in accessibility analyses. All point location data for these facilities were sourced from OSM and the Humanitarian Data Exchange (Humdata) by exporting the data directly in compatible formats such as **GeoJSON** and **Shapefiles**. After the catchment areas were computed, the country was divided into 100 m × 100 m raster cells. This was achieved by defining a grid with a uniform resolution of 100 m, ensuring consistent spatial coverage across the entire study area. The grid was generated using GIS software, aligning with the

country's coordinate reference system to maintain spatial accuracy [126,127]. Next, each cell was assigned the score of the corresponding catchment area. Below is a summary of the scoring systems used for rating access levels to each factor.

4.3.2.1. WTP. Specifically, the analysis of women's travel patterns encompassed access to childcare services, grocery stores, markets, green spaces, and pharmacies. Access to each type of facility was scored as follows: 0–400 m received a 5, 401–800 m received a 4, 801–1200 m received a 3, 1201–1500 m received a 2, 1501–2000 m received a 1, and over 2000 m received a 0. Thresholds were established based on studies that analyzed access to these key facilities and deemed the criteria appropriate [79,128,129].

4.3.2.2. Access to public transport. Locations within 0 to 250 m of public transport stops received a score of 5. Those within 251 to 500 m received a 4, 501 to 750 m received a 3, 751 to 1000 m received a 2, 1001 to 1500 m received a 1, and over 1500 m received a 0. This distance (1500 m) is roughly equivalent to a 15-min walk at a moderate pace [130]. Ivan et al. [93] argue that after five minutes of walking, the willingness to walk to a public transport stop decreases significantly.

4.3.2.3. Access to education and training facilities. Locations within 0 to 2000 m received a score of 5, those within 2001 to 4000 m were assigned a score of 4, locations 4001 to 6000 m away received a score of 3, those 6001 to 8000 m away were given a score of 2, locations 8001 to 10,000 m away were assigned a score of 1, and any location beyond 10,000 m received a score of 0. This analysis was based on Theunynck's [131] recommendations for planning schools. Additionally, it was based on the assumption that, unlike children, older individuals typically commute to higher education institutions [95].

4.3.3. Access to health facilities

the suggested thresholds were adjusted as follows: 0 to 1000 m received a score of 5, 1001 to 2000 m received a 4, 2001 to 3000 m received a 3, 3001 to 4000 m received a 2, 4001 to 5000 m received a 1, and over 5000 m received a 0. This rationale stems from studies conducted by Dotse-Gborgbortsi et al. [132] in Ghana and Zielinski et al. [133] in Östergötland County, Sweden. Both studies noted that willingness to travel to health facilities decreases as distance increases, with a significant decline observed beyond one hour of travel.

4.3.3.1. Access to financial facilities. Based on the study by Bhanot et al. [101], which observed that banks located >3 km away were no longer considered accessible, the thresholds for scoring access to financial facilities were established as follows: 0 to 400 m received a 5, 401 to 800 m received a 4, 801 to 1200 m received a 3, 1201 to 2000 m received a 2, 2001 to 3000 m received a 1, and beyond 3000 m received a 0.

4.3.4. Place-characterization dimension

For the analysis of PD factors, first, the country's territory was divided into 100 m × 100 m raster cells. Similar to the AD factors, the rasterization process involved defining a grid with a uniform 100-m resolution to ensure consistent spatial coverage across the study area. All point location data for PD facilities were sourced from OSM and Mapillary by exporting the data directly in compatible formats such as **GeoJSON** and **Shapefiles**, except for FCV data and Environmental Hazards data. FCV data was extracted from the Armed Conflict Location & Event Data (ACLED) as point-based event records. Meanwhile, Environmental Hazards data was obtained from the Landslide Susceptibility Map and the Global Drought Hazard Project in raster format (GeoTIFF). Below is a summary of the scoring systems used for rating each factor in the PD.

4.3.4.1. Active transport. This factor was evaluated by calculating the average score of four subfactors, sourced from OSM: street crossings, cycle paths, footpaths, and block sizes. These elements are widely recognized in the literature for their role in enhancing active transportation [134–136]. Street crossings, cycle paths, and footpaths were each measured by their availability or presence per raster cell, with scores of 0, 3, and 5 indicating the presence of none, one, or two or more, respectively. Block sizes, indicating encouraged walkability, were scored from 1 to 5 based on their size: 1 for very large block sizes (>1 km), 2 for large block sizes (751 m - 1 km), 3 for moderate block sizes (501 m - 750 m), 4 for small block sizes (251 m - 500 m), and 5 for very small block sizes (<250 m).

4.3.4.2. Safety. The analysis of safety was based on the Nighttime Lights dataset from VIIRS (Visible Infrared Imaging Radiometer Suite). Regions with higher luminosity levels were assigned higher safety scores according to the following scale: areas with luminosity between 0 and 0.05 were given a score of 0; areas with luminosity >0.05, up to 0.25 times the median, received a score of 1; those with luminosity >0.25 times the median, up to 0.5 times the median, received a score of 2; areas with luminosity >0.5 times the median, up to the median, were assigned a score of 3; regions with luminosity greater than the median, up to the 75th percentile, were scored 4; and finally, areas with luminosity greater than the 75th percentile received the highest score of 5. Studies such as those by Mayen Huerta and Utomo [40] in Mexico City, Mexico, and An et al. [137] in Sejong City, South Korea, have established that women feel safer in well-lit areas, highlighting lighting as a key factor in reducing the perception of fear.

4.3.4.3. FCV. The data for assessing FCV events was sourced from ACLED (Armed Conflict Location & Event Data). Using point locations of FCV events, circular buffers with a radius of 5 km were generated to estimate the spatial impact of these events. When the specific impact radius of an event was known, it was used instead. The scoring system was as follows: raster cells overlapping with buffers indicating battles and explosions received a score of 0; raster cells overlapping with buffers indicating explosions and remote violence received a score of 1; raster cells overlapping with buffers indicating violence against civilians received a score of 2; those overlapping with buffers indicating protests and riots received a score of 4; and areas that did not overlap with any event received a score of 5.

4.3.4.4. Education. In the case of Saint Lucia, specific data on the share of STEM university graduates was unavailable at the time of the search. Furthermore, disaggregated data at the district level for any educational attainment outcome was also not available. Hence, we used an estimate from the Central Statistics Office as a proxy, indicating that in 2016, 13–15 % of the female labor force possessed a university degree [138]. As previously mentioned, this percentage was standardized on a scale of 0 to 5.

4.3.4.5. Digital inclusion. No spatially disaggregated information was available for Saint Lucia at the time of the search. However, according to the World Bank's Data Portal, 78 % of the population in Saint Lucia has access to the Internet [125], which translated into a country-level score of 4.

4.3.4.6. Water and sanitation. Water and Sanitation was assessed based on the presence of water points, catch basins, water valves, and fire hydrants. We created 1 km buffers around these points of interest. Raster cells within these buffers were assigned a score of 5, indicating access to water and sanitation, while raster cells outside the buffers received a score of 0.

4.3.4.7. Environmental hazards. Specifically for Saint Lucia, we analyzed the two environmental hazards that have the most significant

impact on women's economic activities on the island: landslides and droughts. Landslide susceptibility was assessed using data from the NASA/GSFC Landslide Susceptibility Map [139], while drought risk was evaluated based on data from the Global Drought Hazard Project, which uses monthly spatial raster datasets from the Climatology and Climate Services Laboratory's Global SPEI Database, covering the period from January 1902 to December 2018 [140]. We reclassified the input data to a standardized scale of 0 to 5 using a linear scaling process, where a score of 5 represented areas with no environmental hazards, and a score of 0 represented areas with the highest level of hazard.

4.4. Practical application

The final score for the level of enablement of women's employment opportunities per raster cell (W), was calculated following the formula below:

In this assessment, each factor within each dimension was given equal weighting. The final calculation of W incorporated equal weights for each dimension, with the two spatially varying dimensions within the country accounting for two-thirds of the final score. This approach ensured a more accurate representation of the spatially oriented analysis. Although the CD factors often exhibit no spatial variation within the country, as policies and laws are typically applied uniformly at the national level, this dimension was given a weight of 1/3. The weighting is justified because, in many countries, cultural norms and traditional gender roles often discourage women from pursuing careers outside the home, creating an unsupportive environment for female workforce participation. These societal expectations can lead to limited opportunities for women, as they are often expected to prioritize domestic responsibilities over professional aspirations [141,142]. Consequently, without laws and regulations to protect women, their access to employment opportunities would be severely restricted. Nonetheless, if a study aims to focus on the spatial variation of **W** within a country, it may opt to exclude **CD** and use the following equation instead:

Importantly, to ensure this research was both replicable and scalable, we developed a QGIS plugin, which is available in the following GitHub repository: https://worldbank.github.io/GEEST/README.html. This plugin allows users to input the necessary data for evaluating each factor, automating the analyses to produce the final scores and maps.

To address the potential mismatch between enabling environments and where women actually live, we used female population density data from WorldPop, categorizing it into three subclasses: low, medium, and high. By combining these population density classes with the enablement classification, 15 distinct enablement-population categories were created. These categories allowed for a better understanding of the spatial distribution of supportive environments relative to where women reside. From a policy perspective, this approach is more efficient in identifying priority areas for intervention by focusing on regions where high population density coincides with lower levels of enablement.

4.4.1. Practical application example

To assess the real-world applicability of the GeoWEAF, we focused on analyzing the enablement scores around the solar plant in La Tourney, Vieux Fort. This analysis aimed to evaluate how effectively the surrounding environment of the plant supports women's access to employment opportunities. By examining the area's characteristics, we identified both the strengths and challenges in promoting gender-inclusive employment in the area.

4.4.2. Data quality assessment

To ensure the adequacy of the spatial data used in the assessment, we conducted a thorough data quality evaluation. The quality assessment focused on evaluating the spatial data of Saint Lucia within OSM using the Ohsome Quality API. The analysis examined the completeness, accuracy, and timeliness of mapped features such as buildings and roads. Results indicated that approximately 70 % of buildings and 90 % of roads have been mapped, with a high degree of completeness when compared to external datasets like Microsoft Global ML Buildings, covering 93 % of buildings and 95 % of roads. The evaluation also explored various accessibility factors, showing that while some facilities like transportation, health facilities, and grocery stores are well-mapped, other facilities such as kindergartens and water sanitation points have lower data reliability.

In addition, the assessment included a Mapillary analysis to evaluate the visual coverage of Saint Lucia's road network. This analysis revealed that major roads are well-covered by Mapillary images, with 92.6 % coverage, but significant gaps exist for minor roads and pedestrian pathways, which are only covered at 62.9 % and 17 %, respectively. These findings underscore the strengths in the existing OSM and Mapillary data while highlighting areas that require further attention to improve the overall quality and coverage, particularly in less-represented infrastructure and categories.

5. Results

The CD, the only dimension without spatial variability across the country, received a score of 4.1, negatively impacted by lower scores in the Parenthood indicator (2), which contributes to the Regulatory Frameworks factor, and the Entrepreneurship indicator (3.75), which contributes to the Financial Inclusion factor.

In terms of the AD, the results for each factor are shown in Fig. 2. Additionally, Fig. 5 displays the overall results for the Accessibility Dimension. As anticipated, the highest levels of access to each factor are concentrated in urban centers.

Fig. 3 presents the results of the PD factors analyses. Notably, the analyses of factors such as Active Transport, Education, and Water and Sanitation reveal some areas where these elements fail to be even moderately supportive of women's access to employment. In addition, Fig. 5 presents the overall scores for Place Characterization.

The overall enablement scores for Saint Lucia (Fig. 4) indicate that, while certain areas in the northwest and south may be considered somewhat enabling, the country as a whole lacks regions that are highly conducive to supporting women's access to employment opportunities. Moreover, the area surrounding the solar plant in Vieux Fort is considered a moderately enabling environment for women's job access, although it does not qualify as having the highest level of enablement.

When we combined the enablement scores results with data on the distribution of women of working age—specifically those aged 35 to 39—across low, medium, and high population densities, we found that the most enabling areas in the island also corresponded with regions of high female population (see Fig. 5). However, we also identified a few areas with very high population density but significantly low enablement scores. For instance, the area surrounding the solar plant in Vieux

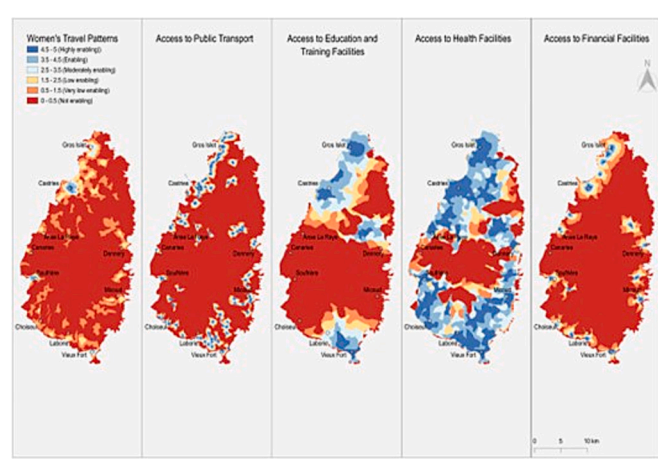

Fig. 2. Results of accessibility dimension factors.

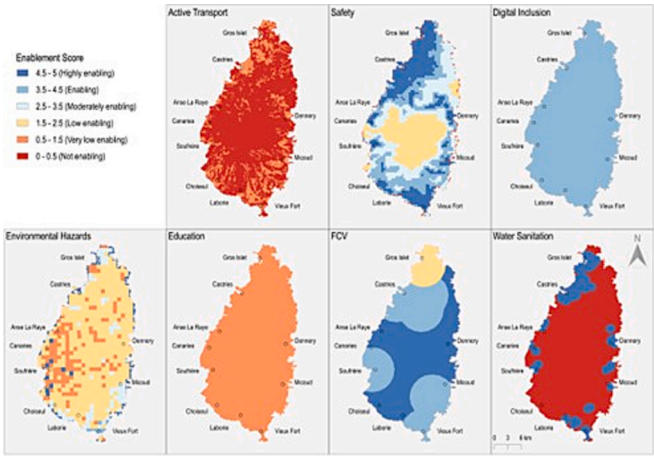

Fig. 3. Results of Place Characterization Factors.

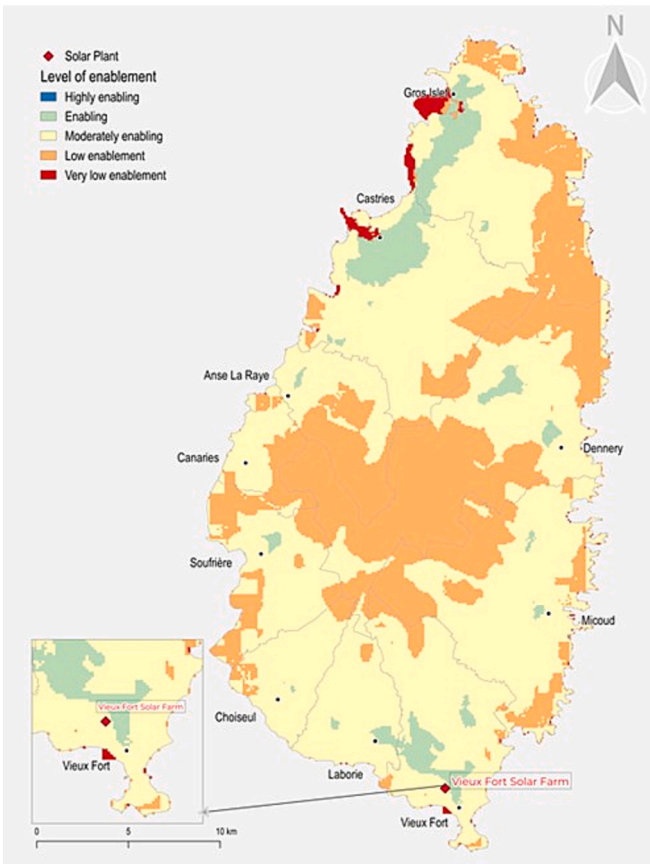

Fig. 4. Overall enablement scores for Saint Lucia.

Fort is characterized by a high female population density but is only moderately enabling.

6. Discussion

This research led to the development of the GeoWEAF, a framework designed to evaluate the environments in which women live, commute, and work, assessing how these environments either facilitate or hinder their integration into labor markets, particularly within the RE sector. To the best of the authors' knowledge, this is the first study focused on understanding the spatial factors that influence women's access to jobs

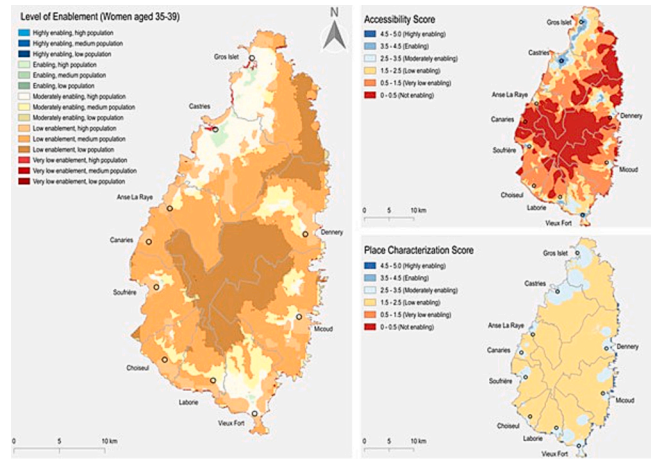

Fig. 5. Enabling scores for Saint Lucia based on female population density, ages 35–39 (left), Accessibility Dimension overall scores (upper-right), Place-Characterization Dimension overall scores (lower-right).

in the RE sector. Similar studies that aim to understand the factors contributing to women's participation in RE often overlook location-specific characteristics that affect job accessibility. For instance, Antasya and Kersana [11], who investigated women's marginalization in Indonesia's RE sector, identified cultural norms and traditional gender roles as the primary barriers to job access. These norms often confine women to caregiving and household responsibilities, making it culturally unacceptable for them to pursue careers in the sector. They also identified societal stigma and stereotypes, a lack of supportive policies, and limited educational and training opportunities as significant barriers [11]. However, the authors do not consider critical spatial factors, such as access to basic services like transportation, healthcare, and childcare, which are essential for enabling women to fully participate in the labor force.

Likewise, the study of Baruah [2], which examines the opportunities and constraints faced by low-income women in accessing employment in India's RE sector, identifies inadequate purchasing power, low social status, and societal gender norms as key barriers to entry. However, it does not consider how the characteristics of specific locations can also hinder women's participation in the sector. By incorporating spatial factors, the GeoWEAF provides a more comprehensive and nuanced understanding of these challenges, supporting the development of targeted interventions that promote inclusive economic growth in the RE sector.As Larsson [143] argues, environments are not gender neutral. Therefore, mainstreaming equal opportunities requires understanding the gendered perceptions of space and accommodating women's needs accordingly [40,144].

Unfortunately, as observed in our literature review, spatial characteristics are often overlooked when attempting to understand women's choices and preferences [36]. As a result, open-source official gender-specific spatial data or sex disaggregated statistics are uncommon in most countries [145–147]. To overcome this barrier, crowdsourced geospatial information from platforms like OSM and Mapillary can provide valuable insights into environmental characteristics relevant to women's mobility and opportunities [148]. Nonetheless, because these sources are contributed by volunteers, they exhibit heterogeneous and diverse quality [149]. This variability stems from the use of various collection methods and the participation of individuals with differing motivations and preferences, which can result in inconsistencies and potential inaccuracies in the data, ultimately affecting the reliability of the results [149].

Despite these data quality constraints in Saint Lucia, the country where the GeoWEAF was applied, the results yield important insights that highlight the spatial dimensions of disparity. Notably, the highest

levels of enablement in the country are not optimal. This finding suggests that improving specific aspects of the environment in the densely populated area near Vieux Fort—such as safety, walkability, STEM education, and access to childcare services—could foster better RE work opportunities for women in the area and promote greater workforce inclusion.

Concerning the CD, the findings suggest that mothers in Saint Lucia still face significant disadvantages, with the Parenthood national score being only 2 (low enabling) [125]. Several studies have concluded that having children carries a significant penalty for women in most low-and-middle income countries, impacting their career progression, earning potential, and perceived competence in the workplace [150–152]. For instance, research by Correll et al. [153] investigating status-based discrimination found that working mothers faced penalties regarding perceived competence and salary. As laws and regulations to protect working mothers are not fully established in Saint Lucia, these issues are adversely affecting the country's CD, which in turn diminishes the support the environment provides to women in the workforce.

With respect to the AD, access to most factors in this dimension is concentrated in the northern and southern peripheries of the island, where population levels are highest. The capital, Castries, is located in the north, while the south features a major hub for industry and transportation near Hewanorra International Airport Of the five factors analyzed, WTP showed the lowest levels of access. This could be due to incomplete data, as our data quality assessment revealed some inconsistencies, or it may result from a lack of urban planning tailored to women's needs. In particular, the WTP factor addresses the caregiving responsibilities of women and highlights how gender-blind urban planning can undermine this crucial role [154]. Neglecting to consider the unique social and economic needs and experiences of women often leads to their exclusion from economic opportunities and decision-making positions [155,156]. Therefore, infrastructure should incorporate these gender-specific requirements, considering the daily realities and barriers women face, including the dual responsibilities of work and caregiving [157,158].

The results in the PD revealed a particularly concerning situation regarding the availability of active transport, a crucial element in ensuring women's right to the city [159]. This issue could be attributed to the limited availability of spatial data concerning active transport features in Saint Lucia. However, additional analysis is needed, as hostile or non-walkable areas can effectively exclude women from fully participating in and benefiting from urban life [40,105,160]. Similarly, the percentage of the female workforce with a university degree is notably low, reflecting less favorable attitudes toward women in the workforce. Barriers to university education experienced by female students limit their advancement into higher-paying positions and contribute to a more pronounced gender disparity in the workforce, particularly in leadership and specialized roles [161].

In particular, factors in the PD, such as Safety, FCV and Water and Sanitation, have been associated not only with increased job access but also with improved quality of life and better well-being outcomes for women [45,121,144,162]. Therefore, addressing these issues is crucial not only for fostering inclusive environments that empower women but also for enhancing their overall well-being and quality of life. This, in turn, can lead to greater social equity and more resilient communities, ultimately contributing to sustainable economic growth and development.

Finally, the analysis of the enablement results for the area surrounding the solar plant at La Tourney in Vieux Fort revealed a moderately enabling environment, though several aspects require improvement. Active transport infrastructure is suboptimal, and higher educational attainment remains a significant concern. These issues are often linked to broader inequitable policy decisions related to gender [163]. Both mobility and education are critical factors in ensuring women's full participation in economic and social life [164]. Furthermore, without reliable, safe, and inclusive transportation options, women are often restricted in their ability to access employment, education, and other essential services [165,166]. This limitation not only reinforces existing gender inequalities but also hinders broader economic development.

Notably, the GeoWEAF's versatility allows for similar analyses to be conducted across various geographic scales and potentially in other industries. However, further research is needed, as RE jobs come with distinct requirements and challenges. Unlike sectors such as agriculture, where access to land and markets are primary concerns [167], the renewable energy sector demands technical skills, specialized training, access to financing, and supportive legal frameworks that promote gender equality [2,6,12]. Moreover, the framework takes into account barriers such as societal attitudes toward women working in male-dominated fields—barriers that can result in harassment, discrimination, and limited participation [168]. These challenges are often less pronounced in sectors like tourism or service industries, highlighting the importance of context when analyzing employment opportunities through this framework. Ultimately, by applying the GeoWAF across different locations, policymakers and researchers can identify the unique spatial challenges and opportunities that influence women's empowerment in RE jobs, enhancing existing analyses. These insights can support the development of more targeted interventions, promoting gender equity and fostering inclusive economic growth in low- and middle-income countries.

7. Conclusion

This study underscores the vital role spatial factors play in shaping women's access to employment opportunities, particularly within the RE sector in low- and middle-income countries. By developing and applying a framework centered on spatial factors (GeoWEAF), the research highlights the significance of addressing geographical disparities and environmental characteristics to close the gender gap in employment. The application of the GeoWEAF in Saint Lucia reveals that, despite some areas of progress, significant challenges persist, particularly concerning walkability, and access to essential services like childcare, pharmacies, and grocery stores—services crucial for women balancing caregiving responsibilities. These barriers significantly hinder women's full participation in the workforce.

Additionally, the limited availability and reliability of spatial data in Saint Lucia further complicate efforts to accurately assess how supportive environments are for women's economic empowerment. Considering spatial factors in evaluating employment opportunities is essential for promoting gender equality in the workforce, especially in sectors like RE, which are critical for sustainable development. Integrating the findings of this framework into policy-making and urban planning can aid stakeholders to better understand and mitigate the barriers women face in accessing jobs in the sector, leading to more targeted and effective interventions. Ultimately, this research calls for a comprehensive, gender-sensitive approach to labor market policies, ensuring that all women, regardless of their location, have equal opportunities to contribute to and benefit from the global transition to renewable energy.

CRediT authorship contribution statement

Carolina Mayen Huerta: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Clara Ivanescu:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis, Data curation. **Benjamin Herfort:** Formal analysis, Data curation. **Piero Badowski:** Formal analysis, Data curation. **Dragos Gontariu:** Visualization, Formal analysis.

Funding

This research was funded by the Canada Clean Energy and Forests Climate Facility through the World Bank Group project titled "Geospatial Assessment of Women's Employment Opportunities in the Renewable Energy Sector."

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Carolina Mayen Huerta reports financial support was provided by World Bank. Carolina Mayen Huerta reports a relationship with World Bank Group that includes: consulting or advisory and employment. Clara Ivanescu reports a relationship with World Bank Group that includes: employment. Dragos Gontariu reports a relationship with World Bank Group that includes: consulting or advisory. No other conflicts of interest. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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